CertJudge Revision Memo: Notation, Claims, Data, and the ICLR 2027 Plan

13 August 2026

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Purpose. This memo is the source of truth for revising *Certifying the Judge* into an ICLR 2027 submission. It fixes the final notation and terminology, states which claims the paper may make and which it must retract, records the audited facts about the human-annotation data, and allocates content to the 9-page main text versus the appendix. Claude Code should implement the paper edits directly from this document (companion prompt: `agent-prompt-iclr2027-port.md`). The full working archive (derivations, handwritten-page transcriptions, reconciliation log) remains in `certifying-the-judge-COMPLETE.*` for provenance; nothing there overrides this memo.

Venue facts (verified 13 Aug 2026). ICLR 2027: abstract deadline **Sep 18, 2026 AoE** (abstracts must be genuine; used for reviewer bidding); paper deadline **Sep 25, 2026 AoE**; submissions via OpenReview. Main text **9 pages** at submission (10 at rebuttal/camera-ready), strictly enforced; references uncounted; unlimited appendix after references, which reviewers are not required to read. Double-blind — identity leaks in main text *or* supplementary are desk-rejected; cite our own prior versions in third person. A short **AI-use statement is required** (uncounted); reproducibility and ethics statements are recommended (uncounted). Style files: `https://media.iclr.cc/Conferences/ICLR2027/iclr-2027-style-files.zip`. Authors on 3+ submissions owe 6 reviews each.

1 Terminology and notation (final)

We separate three objects that earlier drafts collapsed into one scalar. The role labels follow the validity-centered evaluation framework of Salaudeen et al. (2025), which the paper already cites.

Object	Role	Notation	Needs human labels
Trustworthiness (the predicate we care about)	construct	$\mathrm{TW}_m(aj)$	—

Object	Role	Notation	Needs human labels
Trust Index (the construct's measured scalar on a specified panel and item set)	criterion	$\mathrm{TI}_m(aj)$	yes
Proxy Trust Index (label-free behavioral aggregate)	instrument / surrogate	$\mathrm{PTI}_p(aj)$	no

Subscripts follow the L_p convention: the capital letter names the family, the subscript selects the variant. m indexes the **matching criterion** on the target side; $p \subseteq \{1, \dots, K\}$ indexes the **property set** on the proxy side and is **locked before any bridge analysis** (choosing p to maximize the bridge correlation is model selection over $2^K - 1$ subsets and reintroduces the selection circularity).

The m-ladder. m_0: distributional match, $P_\theta(s \mid \cdot, aj) \approx P_H(s \mid \cdot, h_j)$. m_1: correlational, $\mathrm{Corr}(S_{aj}, S_{h_j})$ high over items. m_2: m_1 plus stability, $\mathrm{std}(\hat{s}_{aj})$ low. We write $\mathrm{TW}_2 \equiv \mathrm{TW}_{m_2}$; m_2 is the working target.

Core definitions.

$$\mathrm{TW}_m(aj): \quad \forall a \in L_A, \text{ if } aj \approx_\Pi h_j \text{ then } \big\| P_\theta(s \mid x, y^*, \hat{y}{=}a(x), aj) \;\approx_m\; P_H(s \mid x, y^*, \hat{y}{=}a(x), h_j) \big\|_m \text{ small},$$

where $aj \approx_\Pi h_j$ is the persona precondition (the judge is conditioned toward the rater persona; without it the definition is unsatisfiable over a heterogeneous pool). Items are what the m_1/m_2 criteria integrate over — they are never universally quantified in a correlational statement.

Scores, candidate pools, and datasets. A judge is identified with its scoring function, and the leaderboard is an operator that turns a benchmark subset into a pool of scored candidates:

$$\hat{s}_{aj}(x, y^*, \hat{y}) \;:=\; aj(x, y^*, \hat{y}), \qquad L_A(b') \;:=\; \{\, (x,\, y^*,\, \hat{y}_a(x)) \;:\; (x, y^*) \in b',\; a \in L_A \,\}.$$

$$D_{\hat{S}_{aj}|b', L_A} = \{\hat{s}_{aj}(z) : z \in L_A(b')\}, \qquad D_{\hat{S}^*_H|b', L_A} = \{s^*_H(z) : z \in L_A(b')\}.$$

Both datasets are indexed by the *same* set $L_A(b')$, so **pairing is a typing fact rather than a side condition**: a correlation between them is well-formed precisely because they are functions on a common index. Repeats are aggregated before this point — K_a agent generations, K_{aj} judge repeats and K_h human passes are three distinct indices that never align pairwise.

$$\mathrm{TI}_{m_1}(b', aj, L_A, H) \;=\; \mathrm{Corr}(D_{\hat{S}_{aj}|b', L_A},\; D_{\hat{S}^*_H|b', L_A}) \qquad \mathrm{PTI}_p(aj) \;=\; \mathrm{GM}(P_k(aj) : k \in p)$$

$$\mathrm{TI}_{2'}(b', aj, L_A, H) \;=\; \Big(\mathrm{clip}_{[0,1]}(\mathrm{Corr}(D_{\hat{S}_{aj}|b', L_A},\, D_{\hat{S}^*_H|b', L_A})) \cdot (1 - \mathrm{STD}[D_{\hat{S}_{aj}|b', L_A}])\Big)^{1/2}$$

Three standing corrections to watch when transcribing from working notes into the manuscript: the stability term is **judge-side** (P_4, the label-free conjunct); it is **orientation-flipped** ($1 - \mathrm{STD}$, since raw STD would reward instability); and the correlation is **clipped to** $[0, 1]$ before entering the geometric mean.

Predicate versus scalar. TI_m is the scalar; TW_m is the predicate it induces, at a stated threshold:

$$\mathrm{TW}_m(b', aj, L_A, H) \;:=\; [\,\mathrm{TI}_m(b', aj, L_A, H) \;\geq\; \tau\,].$$

Working notes that write $\mathrm{TW}_{2'}(\cdot) = (\mathrm{Corr} \cdot \mathrm{STD})^{1/2}$ are defining the scalar and should be read as $\mathrm{TI}_{2'}$. The judge-synthesis goal is stated on the same footing: generate aj such that $\hat{S}_{aj|b,L_A} \approx_{\mathrm{TW}_2} \hat{S}^*_{H|b,L_A}$.

The property suite, and the argument-slot swap that is the paper's thesis. The suite is

$$\mathbf{P} = \{P_{\mathrm{rfl}},\ P_{\mathrm{mb}},\ P_{\mathrm{mms}},\ P_{\alpha SI}\} \quad \text{(identity; bug monotonicity; missing-spec monotonicity; } \alpha\text{-semantic invariance)},$$

$$\mathrm{PTI}_p(aj) \ = \ \Big[\prod_{k\in p} P_k(aj)\Big]^{1/|p|}.$$

Identity is stated over **all three argument slots**, not just the candidate: for identical inputs (x, x), $(\hat{y}_a, \hat{y}_a)$ and (y^*, y^*) the judge should return ≈ 1, and the check may also be run on deliberately corrupted files (if two files are identical the judge should say so, however bad they are). Every P_k maps to $[0, 1]$ with 1 conforming; for the monotone properties the map is $\frac{1-\rho}{2}$ — never $|\rho|$, which would also reward a maximally anti-monotone judge ($\rho \to +1$, scoring buggier artifacts *higher*), and never bare $1-\rho$, whose range $[0, 2]$ lets the aggregate exceed 1. Orientation and range errors of this family — raw STD instead of $1 - \mathrm{STD}$, $|\rho|$ or $1-\rho$ instead of $\frac{1-\rho}{2}$, unnormalized products instead of geometric means — are the most frequent defect in working drafts of these formulas and belong on the manuscript check-list.

One operator, two ground truths. Define a validity–stability functional on paired vectors, and its correlation-only component:

$$\mathcal{M}^C(D_1, D_2) \ = \ \mathrm{clip}_{[0,1]}(\mathrm{Corr}(D_1, D_2)), \qquad \mathcal{M}^{VS}(D_1, D_2) \ = \ \big(\mathcal{M}^C(D_1, D_2) \cdot (1 - \mathrm{std}(D_1))\big)^{1/2}.$$

The criterion and the properties are then **the same measurement against different ground truth**:

$$\mathrm{TI}_{2'}(aj) = \mathcal{M}^{VS}(\hat{S}_{aj},\ \hat{S}^*_H), \qquad P_{\mathrm{mb}}(aj, y) = \mathcal{M}^C(\hat{S}_{aj}\restriction_{\mathrm{ladder}(y)},\ \text{bug level}) \text{ (orientation-flipped)}.$$

This is the $\mathbf{H} \to \mathbf{P}$ swap realized at the operator level: one functional, one argument changed — an expensive human panel, or a cheap constructed ladder. It also answers directly the question a reviewer will ask first, *why should behavioral properties predict human agreement at all?* They are not analogous to the criterion; they are the criterion's own operator applied to synthetic ground truth.

Two shapes, not one. $\mathcal{M}^{VS}(D_1, D_2)$ requires a meaningful second argument. The *ladder* properties have one — the rung index — so P_{mb} and P_{mms} take the correlational form. The *invariance* properties do not: for P_{rfl} and $P_{\alpha SI}$ the second argument is constant, correlation is undefined, and the right functional is a deviation measure, $1 - \mathbb{E}|\Delta s|$. The unification is real but covers two of the four properties.

Per-item aggregation. Each item yields one validity–stability scatter (judge scores against rung), $\mathcal{M}^{VS}$ maps that scatter to a scalar, and the property is the geometric mean over items:

$$P^{\mathrm{GM}}_{\mathrm{mb}}(aj, P_{\mathrm{mb}};\, D, \mathrm{GM}) \ = \ \Big(\prod_{y\in D} \mathcal{M}^{VS}(\hat{S}_{aj,y,\mathrm{bugs}},\ \mathrm{bugs})\Big)^{1/|D|}.$$

Which variance. Inside a ladder property the stability component is **repeat variance at a fixed rung**, averaged across rungs — never the spread of scores *along* the ladder. Spread along the ladder should be large for a good judge, since scores must move as bugs accumulate; penalising it would contradict the correlation requirement in the same expression.

Variance appears on exactly one side of the bridge. Instantiating $\mathcal{M}^{VS}$ on both sides would put a judge-level repeat-noise factor into both scalars, making part of the bridge correlation mechanical rather than evidential — a high-variance judge scores low on both axes whatever the properties do, which at $n \approx 10$ judges could carry the entire trend. Two coherent configurations:

	Target	Properties
Config A	$\mathrm{TI}_{2'} = (\rho \cdot (1-\sigma))^{1/2}$	$\mathcal{M}^C$, correlation only
Config B (adopted)	TI_{m_1}, human correlation only	$\mathcal{M}^{VS}$ per item, correlation and stability

Config B is the working choice, for two reasons: it matches the unified $\mathcal{M}^{VS}$ formulation, and it matches the existing implementation, where the label-free index already carries variance penalties while the gold quantity is plain human correlation. Config A would require rewriting both sides. $\mathrm{TI}_{2'}$ retains its role as the deployment/decision criterion. Report $\mathrm{Corr}(\sigma_{\text{target}}, \sigma_{\text{proxy}})$ once as a collinearity diagnostic regardless of configuration.

Argument conventions. The paper uses a **semicolon**: varying arguments precede it, fixed context follows — $\mathrm{TW}_{2'}(aj, H;\, b, L_A)$, $\mathrm{PTI}_{2'}(aj, \mathbf{P};\, L_A, b)$ — and the aggregator is an explicit argument rather than baked in: $P_k^{\mathrm{Agg}}(aj, P_k;\, D, \mathrm{Agg})$, defaulting to Agg = GM. Working notes use a conditioning bar for the same split because it is easier to read at speed; the bar must be translated to a semicolon on the way into the manuscript, since | carries probabilistic connotations these arguments do not have.

Property estimators (two-level geometric aggregation). Each property aggregates over items and each is oriented so that 1 is conforming. Both aggregation levels are geometric means — within a property over items, and across properties over the suite:

Property	Estimator	Ladder / transform
P_{rfl} identity	$\big(\prod_{y \in D} \hat{s}_{aj}(y, y)\big)^{1/\lvert D\rvert}$	none (self-pairs; may also be applied to perturbed files)
P_{mb} bug monotonicity	$\big(\prod_{y \in D} \frac{1-\rho_y^{\mathrm{bug}}}{2}\big)^{1/\lvert D\rvert}$, $\rho_y^{\mathrm{bug}} = \mathrm{Corr}_i(\hat{s}_{aj}(b_i(y)),\, i)$	cumulative bug ladder $b_0, \ldots, b_{\#\mathrm{bugs}}$
P_{mms} spec monotonicity	identical shape, ρ_y^{spec}	cumulative specification-removal ladder
$P_{\alpha SI}$ semantic invariance	$\big(\prod_{y \in D} (1 - \lvert\hat{s}_{aj}(x, y^*, T_\alpha(\hat{y})) - \hat{s}_{aj}(x, y^*, \hat{y})\rvert)\big)^{1/\lvert D\rvert}$	α-renaming of bound variables

Why the $1/|D|$ exponent is not cosmetic. A raw product is scale-dependent: 72 items at 0.99 give 0.48, while the same judge on the full benchmark's 471 candidates gives $0.99^{471} \approx 0.009$. Since PTI is computed on b and TI on $b' \subseteq b$, raw products would make the two sides of the bridge incomparable **by construction** rather than by judge behavior. The geometric mean is scale-free, stays in $[0, 1]$, and preserves the zero-annihilating soft-AND semantics. Likewise $1 - \rho$ ranges over $[0, 2]$ and must be $\frac{1-\rho}{2}$ before aggregation.

Stability appears once on each side, measuring different things. The $(1 - \mathrm{std})$ conjunct of $\mathrm{TI}_{2'}$ is *variance under resampling the judge* (repeats, prompt paraphrase); $P_{\alpha SI}$ is *sensitivity to rewriting the input* under a semantics-preserving transform. These are distinct measurements, so both may appear without double-counting — provided the target's term is computed from repeats only, the property from rewrites only, and (per the operator rule above) no $\mathcal{M}^{VS}$ variance factor is smuggled into the property estimators. This supersedes the earlier proposal to drop stability from the bridge aggregate, and it settles the suite at $K = 4$.

The thesis is then a single substitution in one argument slot:

$$\mathrm{PTI}(b,\, L_A,\, aj,\, \mathbf{P}) \quad \approx_C^{AJ} \quad \mathrm{TI}_{2'}(b',\, L_A,\, aj,\, \mathbf{H}), \qquad b' \subseteq b.$$

The property suite occupies exactly the slot the human pool occupies; that swap *is* the amortization claim, and the rest of the protocol exists to license it. Two indices must not be confused: the $\approx_m$ inside TW_m

correlates scores **across items**, whereas $\approx_C^{AJ}$ correlates the two scalars **across judge configurations** — one level up, different sample size, and the source of the outer power constraint.

Effective parameters and the amortization asymmetry. The gold criterion is only measurable on the annotated subset $b' \subseteq b$; the proxy runs on the full benchmark:

$$\mathrm{PTI}(b,\, L_A,\, aj,\, P) \;\approx\; \mathrm{TI}_{2'}(b',\, L_A,\, aj,\, H), \qquad b' \subseteq b.$$

The design statement that follows: *select a subset $b' \subseteq b$, run the leaderboard L_A on it to produce $L_A(b')$, collect human scores on that pool, and ask whether the judge's scores match the humans' in the* TW_2 *sense.*

This exposes four generalization axes, all of which the paper must name: held-out **judges** (the primary claim), new **agents**, new **benchmarks**, and within-benchmark **item-level extrapolation** $b' \to b$ (the bridge is validated on 72 items and the proxy is used on 157×3 candidates).

Rename ledger (apply in one pass across text, figures, captions, and code identifiers):

Old	New
$\mathrm{TI}_{\mathrm{var}}$, "trust index", "cert index"	PTI_p
Align, "human alignment/Spearman" (per judge)	TI_{m_1}
"certification protocol", CERTIFY	**amortized judge validation**, SCREEN
pass1 / pass2	**strict / lenient** (the pass names are inverted vs. chronology; see §4)
"reflexivity" (lay contexts)	**identity** (the paper's existing term)
$\mathrm{TW}_{2'}(\cdot) = (\mathrm{Corr} \cdot \mathrm{STD})^{1/2}$ in notes	$\mathrm{TI}_{2'}$ (scalar); TW is the thresholded predicate
P1/P2/P3/P4	P_{rfl} / P_{mb} / P_{mms} / $P_{\alpha SI}$ (mnemonic subscripts; keep numeric aliases in tables only)

Bans. "Trust Index" never appears in the title or public branding (title collision: arXiv:2507.16028, Rug et al. 2025); TI_m survives as internal math. No `CI` or `CC` abbreviations. No hat notation: a hat is earned by a fitted calibration map $f_{\mathrm{cal}} : \mathbf{P} \mapsto \widehat{\mathrm{TI}}_{m_1}$, which is premature below roughly 40 judge configurations; the raw aggregate never receives one.

2 Claims the paper may make, and the retractions it must reflect

2.1 No "sound rejection by construction" at the ratified target. Under TW_2, correlation is an aggregate: a judge can carry high TI_{m_1} while exhibiting a local bug-ladder reversal or one rename sensitivity, so P_k low does not logically imply TI_{m_1} low. Necessity of P_1–P_3 holds only for pointwise trustworthiness. Two repair routes, both acceptable:

- **Route A (definitional).** $\mathrm{TW}(aj) := [\mathrm{TI}_{m_1} \geq \tau_A] \wedge \bigwedge_k [P_k \geq \tau_k]$; a probe violation then rejects by definition, and "properties predict alignment" is reported as a separate empirical result.
- **Route B (conditional falsification lemma).** If $|J(z) - H(z)| \leq \varepsilon$ pointwise and humans satisfy a margin $H(x) - H(Tx) \geq \delta$ on a transformation T, then $J(x) - J(Tx) \geq \delta - 2\varepsilon$; an observed reversal exceeding $2\varepsilon - \delta$ refutes ε-alignment, and human-invariant T gives $|J(x) - J(Tx)| \leq 2\varepsilon$. The paper sentence: *under stated metamorphic-validity and margin assumptions, sufficiently large probe violations falsify pointwise ε-alignment.* The margins δ are currently assumed, not measured — a small human collection on perturbed ladder items (a few dozen ratings) turns the lemma into a grounded theorem and simultaneously tests whether ALPHA renaming is human-invariant, which is the standing P_4 failure mode.

2.2 Acceptance is licensed, never free. High PTI_p "certifies" (scare quotes deliberate) only because the bridge correlation with TI_{m_1} was demonstrated, and only within the validated judge population. The

abstract must state the screen semantics explicitly; the prior desk rejection traces to an abstract that claimed prediction while the experiments reported a screen that found no qualifying signal.

2.3 The geometric mean is a soft AND. $\mathrm{GM}(1,1,1,0.5) \approx 0.84$: the aggregate collapses only at exactly zero. GM remains the right ranking summary (zero-annihilating, imbalance-penalizing), but the hard screening decision uses per-property confidence bounds — FAIL-SCREEN when $\mathrm{UCB}(\hat{P}_k) < \tau_k$ for any required k — and the auditable object is always the property vector $\mathbf{P}(aj)$, with $\min_k P_k$ reported when worst-case conformance matters.

2.4 One conjunct of TW_2 is directly estimable label-free — the stability conjunct, via P_4 (an estimate with a CI, not a verification). Never "half is free." Consequently P_4 leaves the primary bridge aggregate: the headline bridge is $\mathrm{PTI}_{\{1,2,3\}}$ versus TI_{m_1}, the reported object is the profile ($\mathrm{PTI}_{\{1,2,3\}}$, $\hat{P}_4 \pm$ CI, violation flags), and the full GM survives only as a single-number deployment convenience. Stability must never appear on both sides of the bridge.

2.5 Labels are de-indexed, not eliminated. The honest cost claim:

$$C_{\text{amortized}} = \underbrace{|\Pi|\, n\, c_{\text{human}}}_{\text{persona conditioning}} + \underbrace{|AJ_{\text{val}}|\,|I|\, c_{\text{human}}}_{\text{bridge, once}} + \underbrace{|AJ|\,|A|\,|Q|\, c_{\text{API}}}_{\text{per screened judge}}$$

— $O(1)$ in agents and screened judges, $O(|\Pi|)$ in personas. The criterion side alone costs $O(|H| \cdot |b'| \cdot |L_A|)$ human judgments — one rating per (problem, agent-candidate) triplet per rater — which reproduces the audited count exactly: $1 \times 24 \times 3 \times 2$ rounds $= 144$ analyzed labels. Approved phrasings: "no new human labels per screened judge," "human-calibrated," "human-label amortized."

2.6 Three-claim structure. Claim A (definitional): aj is TW_2-trustworthy — needs humans. Claim B (proxy): $\mathrm{PTI}_p(aj) \geq \tau$ — label-free. Claim C (bridge): $\mathrm{PTI}_p \approx_C \mathrm{TI}_{m_1}$ over held-out judges — established once. At m_2, failing B is evidence against A with strength set by C; refutation requires Route A or Route B. Passing B never establishes A except through C, population-scoped.

3 Two jobs, two validations

The instrument does two jobs that require different evidence. The paper studies both and never conflates their metrics.

	Screen (gate before humans)	Surrogate (predict the criterion)
Claim	catches unacceptable judges cheaply	PTI tracks TI_{m_1} numerically
Decision	FAIL-SCREEN / PASS-SCREEN / DEFER	predicted score with interval
Metrics	false-pass rate, false-reject rate, defer rate, coverage, **human ratings saved at a fixed false-pass tolerance**	held-out rank correlation, MAE/calibration, prediction intervals, transfer across families/prompts/personas/agents

Decision utility is the headline scalable-oversight number: *at false-pass tolerance α, the screen avoids X% of human validations*, with sensitivity to shift and a stated re-validation horizon. For power-limited analyses we pre-register a smallest effect size of interest (SESOI; candidate $\rho_0 = 0.5$, the level below which screening saves less than it costs) and run equivalence tests, so an inconclusive interval is reported as inconclusive and a bounded null becomes a finding. A learned calibration map is deferred until roughly 40 judge configurations exist; at $n \approx 8$–13 the honest object is the pre-specified aggregate with its raw held-out correlation.

Data pipeline: what feeds PTI, what feeds TI, and where the bridge correlation lives. Three datasets with distinct index sets, and one type rule: *items pair inside each side; judges pair across sides.*

1. **Probe-response tensor (PTI's data; label-free).** $D_Q(aj)[f, T, r, k] = \hat{s}_{aj}(T_r(x_f), g_f)$ over base artifacts $f \in F$ (the published run: $|F| = 184$), transform families T (identity-self; RENAME / REORDER / STYLE / ALPHA; bug ladder L0–L4; cumulative spec-removal ladder), rung r, and repeats $k \leq K_{aj} = 3$. This tensor has **no human axis**: the supervision that labels provide on the gold side is provided here by construction — the ladder order is known because we built it. The suite Q is scripted once from existing candidates and golds, cached, and reused across judges; per new judge the cost is inference only (order $|F| \times$ (variants+rungs) $\times K_{aj}$ API calls, zero annotation).
2. **Panel scores (TI's judge side).** $D_{\hat{S}_{aj}|b',L_A}$: the judge's repeat-aggregated scores on the 72 aligned pairs of $L_A(b')$ — required only for judges in the bridge set, never for newly screened judges.
3. **Human tensor.** $D_{\hat{S}^*_H|b',L_A}$: the same 72 candidates $\times$ {strict, lenient, avg}; fixed, collected once.

Estimators collapse (1) to the property vector — e.g. $P_2 = \frac{1}{|F|}\sum_f \text{clip}(\frac{1-\rho_f}{2}, 0, 1)$ with ρ_f the Spearman correlation of (rung, median-over-repeats score); P_4 alone consumes the un-aggregated repeat axis — and (2)+(3) collapse to $\text{TI}_{m_1} = \text{Spearman}(D_{\hat{S}_{aj}|b',L_A}, D_{\hat{S}^*_H|b',L_A})$ over the 72 shared candidates, per label view. **The bridge dataset is judge-indexed:** $\{(\text{PTI}_{\{1,2,3\}}(aj_j),\ \text{TI}_{m_1}(aj_j))\}_{j=1}^n$, one point per judge configuration ($n = 8$–13 today, target ≈ 40), with $\hat{\rho} = \text{Spearman}_j$ bootstrapped over judges. There are correspondingly **two scatter levels**, both instances of the same operator: an inner per-file plot of judge score against ladder level (a $\mathcal{M}^C$ scatter, one per $y \in D$), and the outer per-judge bridge plot of PTI against TI. D_Q and $D_{\hat{S}^*_H|b',L_A}$ share no index and are never correlated row-wise — humans never rated perturbed variants, by design. This is why the outer power constraint counts judges, not items, and why the fix is more judge configurations.

4 Data: audited ground truth the paper must state

A repository audit (`cert-judge@main`, 13 Aug 2026; write-up in `experiments/26_human_label_dataset_audit/`, tidy source of truth `results/aligned_panel_n75.tsv`) established the following. The revised paper states these facts plainly.

Fact	Audited value
Annotated panel	**75 rated candidate–gold pairs = 25 problems $\times$ 3 agent generations; 72 analyzed** (24 problems). The submitted text's "24 $\times$ 3 = 75" never multiplies out
The dropped 3	`0_unsafeMultiply_safe` rows carry a pre-renumbering filename, miss the gold lookup, and are skipped (`missing_gold`); they drop rather than add. The three 1.5-scores in the score-dispersion figure are an unrelated duplication (`4_json_decoder`, scored 1 and 2, averaged)
Raters	**One annotator, two rubric-strictness rounds** — not two raters. The published pass names are inverted relative to chronology (sha-verified); rename to strict/lenient. No per-row rater field exists anywhere in the label data
Label totals	**150 rated (75 $\times$ 2), 144 analyzed (72 $\times$ 2)**, 0–5 rubric
Inter-round agreement	Spearman 0.249, Pearson 0.375, QWK 0.372 on the 72. With one rater this is **rubric stability**, not inter-rater agreement: the criterion itself moves this much under a strictness change alone

Fact	Audited value
Item selection	Stratified 5-per-split, **hand-picked** within split, frozen 2026-04-07 (commit `3a7b75c`), no script, no seed, not compile-filtered. Estimand: alignment on a fixed curated panel — say so. A seeded selection template (study 41, seed 41, $n = 30$) exists but was never run; it is the upgrade path if a probability sample is demanded
Agents on the slice	codex / claude / llama (`AGENT_MAP`); the full candidate pool is 471 files = 157 problems $\times 3$ (problem granularity; 857 is the theorem count)
The 5-rater pool	A **separate population**: 343 labels over 78 pairs, 5 raters, revised rubric. It correlates with the calibration panel at only $\rho \approx 0.13$
Control-driven correlation	The published 8-configuration panel includes edit-distance and random controls spanning most of the unit square; a bridge correlation computed with them included can be carried by the controls. Report primaries-only alongside pooled, everywhere
Number hygiene	The draft's MAE/MSE (0.778/1.139) uses first-of-duplicates; the cached mean-of-duplicates values are 0.7708/1.1215 — pick one convention and regenerate. The "approximately 72-75" hedge is determinate: 72

Interpretation the paper needs (one paragraph each). (i) With $|H| = 1$, the panel criterion is one person's judgment under a stated rubric — legitimate as a *scoped* criterion via the $h_j = (\text{person}, \pi)$ decomposition, but it must be said, and the rubric is part of the criterion's identity. (ii) **Criterion drift, then power**: the held-out null is documented in-repo as caused by testing against a different rater pool under a revised rubric ($\rho \approx 0.13$ with the calibration criterion) — a bridge calibrated on criterion A and tested on criterion B with $\text{Corr}(A, B) \approx 0.13$ fails at any judge count. Power remains binding on top: the outer correlation's n is the number of judge configurations (8–13 today; CI roughly ± 0.5), and the fix is more judges (~40, API-priced), not more items. The inner correlation runs over 72 items (CI roughly ± 0.2); report both CIs. (iii) All reported correlations to date are **validation-set numbers** — the same labels tuned the index form, aggregation, and thresholds — and are labeled as calibration evidence; held-out claims come only from the locked protocol.

5 Packaging for ICLR 2027

Title shortlist (no "Trust Index"; graded /10):

1. *The Surrogate Gap: When Do Label-Free Behavioral Tests Track Human Agreement for LLM Judges?* — 9
2. *Do Behavioral Stress Tests Predict Human Agreement? Scalable Validation of LLM Judges for Formal Code* — 9
3. *JudgeProbe: Amortizing Human Validation of LLM Judges with Falsifiable Behavioral Tests* — 8.5
4. *Screen, Then Validate: Scalable Oversight of LLM Judges* — 8
5. *Falsifiable Behavioral Properties for Amortized Validation of LLM Judges* — 7.5 (safest)

("Necessary but Not Sufficient" is withdrawn until Route B is proven or Route A adopted.)

Contributions (survives positive, negative, heterogeneous, or family-restricted outcomes). (1) We distinguish scoped judge trustworthiness TW_m (construct), its human-grounded criterion TI_m measured

on a specified panel, and the automated behavioral instrument $\mathbf{P}(aj)$. (2) We introduce a reusable suite of falsifiable behavioral probes for LLM judges of formal-code artifacts with pre-registered ladders and estimators. (3) We evaluate whether $\mathrm{PTI}_{\{1,2,3\}}$ is a valid surrogate for TI_{m_1} on held-out judge configurations, separating property violations (screen) from prediction (surrogate), with SESOI-anchored equivalence analysis where power is limited. (4) We measure the screen's decision utility: human validations avoided at a fixed false-pass tolerance.

One-sentence thesis. We test whether automatically measured behavioral properties can predict a scoped human-alignment criterion for unseen LLM judges, thereby amortizing — not eliminating — the cost of human validation.

Figure 1 (the bridge figure). One point per judge configuration: $\mathrm{PTI}_{2'}$ (label-free proxy, computed on b) on the x-axis against $\mathrm{TI}_{2'}$ (human criterion, computed on b') on the y-axis, both on $[0, 1]$, with marginal densities, error bars on both axes, the $y = x$ diagonal, and a bootstrap fit band over judges. Two anchors make the scale legible: a uniform-random judge near the origin and a reference good judge near $(0.87, 0.89)$. **Correlations are reported twice — primaries only and with controls included** — because random and edit-distance controls sit near the origin and can carry the whole correlation on their own; presenting only the pooled number invites the obvious reviewer objection that the effect is the negative control. Implementation prompt: `agent-prompt-bridge-plot.md`.

Figure elements map onto the m-ladder. The item-level figure should be read as three diagnostics stacked, one per rung, because $\mathrm{TI}_{2'}$ sits at m_2 and a reader needs the rungs beneath the number:

Element	Shows	Rung
Marginal distributions of judge and human scores, and any divergence between them	calibration gap; range restriction in the panel	m_0
Joint structure, summarised by ρ (and τ_b under ties)	alignment	m_1
Per-item repeat-std error bars	judge self-consistency	m_2

Two consequences. First, **marginals alone cannot identify a judge**: shuffling a perfectly aligned judge's scores across items leaves both marginals unchanged and sends ρ to zero, so marginal agreement is necessary and nowhere near sufficient. The dependence structure — not the two marginals — is the load-bearing object, and ρ is its summary. Second, a nonparametric 2D density is the wrong *estimator* of that structure at $n = 72$ with roughly nine distinct human values: summarise dependence with ρ and a CI, and use count-based encodings (hexbin, size-by-count) for the visual.

Calibration gap as a separate axis (future work). Judges in the current panel compress into roughly $[0, 0.3]$ while humans occupy $[0.6, 0.9]$: severe marginal mismatch alongside moderate alignment. This costs nothing today because TI is correlation-based and therefore level-invariant by construction, which is worth stating explicitly rather than leaving a reader to infer that the offset is a failure. It also names a clean follow-up: report a divergence between the two score distributions (total variation or Wasserstein) as a **calibration gap**, distinct from alignment, and study when calibration begins to matter — the m_0 rung of the ladder, currently unused.

Figure hygiene (from the first item-level plots). Judge and human rubrics are on uncalibrated scales (0–10 vs 0–5), so judges sit well below humans in level while the correlation-based criterion is deliberately invariant to that offset; figures must not imply a calibration target the metric does not claim. Human labels are averages of two integer ratings, so item-level data is a tie-heavy lattice: report Kendall τ_b alongside Spearman, and avoid smooth 2D densities. Always display the criterion's decomposition $(\rho \cdot (1 - \sigma))^{1/2}$ rather than the composite alone — a judge with $\rho = 0.04$ and $\sigma = 0.05$ still scores $\mathrm{TI}_{2'} \approx 0.2$, which is monotone in ρ and so harmless for ranking but overstates the level.

Report correlations against the estimated ceiling, not against 1.0. Inter-pass Spearman on the 72 items is 0.249 (one rater, two rubric strictness levels). Labels are the average of the two passes, so the

reliability of the averaged criterion is roughly $2(0.249)/(1+0.249) \approx 0.40$ by Spearman–Brown, and the attenuation ceiling on any judge–human correlation is about $\sqrt{0.40} \approx 0.63$. A judge at $\rho = 0.48$ is therefore near 76% of the achievable maximum, which is a materially different claim from "0.48 out of 1.0." Two caveats belong in the text: Spearman–Brown assumes parallel forms, and the two passes are different rubrics rather than replications; and with a single rater the ceiling conflates rubric sensitivity with rater noise. Both caveats point the same way, so the ceiling is a lower bound on what is achievable and should be reported as such.

Range restriction in the panel. Human scores concentrate in 0.6–0.9 with almost no mass below 0.5, which attenuates every correlation computed on this panel. This is a property of the item selection – hand-picked without controlling for quality spread – not of the judges. The next collection should select items spanning the quality range, and the current figures should show the marginal distributions so the restriction is visible rather than implicit.

Nine-page allocation.

Main text (9 pp)	Appendix (unlimited; reviewers not obliged to read)
Problem + amortization framing; construct/criterion/instrument; definitions (m-ladder compressed to one table); properties and estimators; screen-vs-surrogate protocol; bridge results with CIs and validation/test labeling; decision-utility result; criterion-drift and single-rater disclosure; limitations	Full m-ladder derivation; Route B lemma proof and δ-measurement design; index-formulation comparisons (v1/v2/GM/PTI); per-property diagnostics and failure modes (ALPHA analysis, reversal rates); audit details and counts reconciliation; prompt templates; threshold-calibration sweeps; expanded judge-panel analyses; pre-registration documents

Compliance items. Genuine abstract by Sep 18 (bidding uses it). Anonymize: strip acknowledgments and identifying repo links (use an anonymized repo or zip), cite the workshop version in third person. Include the required AI-use statement and a reproducibility statement (both uncounted). Ethics statement only if warranted.

6 Action items

1. **72-vs-75, Option B (recommended):** rename the three stale label rows to `1_unsafeMultiply_safe` and regenerate the affected experiment outputs, making $n = 75$ literally true — near-zero marginal cost since a full single-artifact regeneration of all headline numbers is already committed. Fallback Option A: reword to "75 rated from 25 problems; 72 analyzed (stale filename)."
2. **Rename strict/lenient** everywhere (paper, code, cached results).
3. **Commit the audit folder** (currently local-only) so its pointers resolve for the team.
4. **Re-run the audit scripts before editing numbers in the manuscript**; agent-reported figures get the same verification discipline as any others.
5. **Expand judge configurations toward ~40** for outer-bridge power; pre-register the SESOI and the locked held-out protocol (single rubric, single pool per claim).
6. **Run the δ-measurement collection** (human ratings on perturbed ladder items) to ground Route B and adjudicate ALPHA human-invariance.
7. ~~Decide whether $P_{\text{ps\&SI}}$ splits.~~ **Settled:** the fourth property is $P_{\alpha SI}$ (stability under α-semantic invariance), and repeat/prompt variance stays in the target's $(1 - \text{std})$ conjunct. $K = 4$; p can be locked.
8. **Orientation check across the manuscript and code:** every property and stability term oriented so that larger is better ($1 - \text{STD}$; $\frac{1-\rho}{2}$, not $|\rho|$), every correlation clipped to $[0, 1]$ before entering a geometric mean, and the bridge correlation indexed over judges rather than items.
9. **Build the bridge figure** (`agent-prompt-bridge-plot.md`) and, in the same pass, recompute the headline bridge correlations **with and without the negative controls**; if the primaries-only correlation is materially weaker, that is a finding to report, not a number to suppress.

10. Open notational threads: the "IS" benchmark-subset label is unidentified; the prompt dimension k_x is deliberately deferred; one subscript-discipline pass over the manuscript (m on the gold side, p on the proxy side).